Exercise 1 - Computational Statistical Physics

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Physicsc MSc/ 25-935-701 8/3/2026

Introduction to Exercise 1

The Monte Carlo simulation of the two-dimensional Ising model on a square lattice was performed using the single-spin flip Metropolis algorithm. For each temperature, the system was first equilibrated and then measurements were taken over a sequence of sampled configurations. The average energy per spin $\langle E \rangle / N$, the absolute magnetization per spin $\langle |M| \rangle / N$, the magnetic susceptibility χ, and the heat capacity C_V were computed as functions of temperature.

Task 1: Measurement of Physical Observables

The code implements a Monte Carlo simulation of the two-dimensional Ising model on a square lattice of size $L \times L$ using the Metropolis algorithm. Initially, a random spin configuration is generated, where each spin takes the value $+1$ or -1. For each value of the inverse temperature β, the initial total magnetization and total energy of the system are computed.

For every temperature, the simulation consists of two main stages. First, a thermalization stage is performed in order for the system to reach thermodynamic equilibrium. During this stage a large number of Monte Carlo moves are executed without recording measurements. Each move randomly selects a spin on the lattice, computes the energy change ΔE that would result from flipping that spin, and decides whether to accept the flip according to the Metropolis criterion. If $\Delta E \leq 0$, the flip is always accepted, while if $\Delta E > 0$ it is accepted with probability $e^{-\beta \Delta E}$.

After thermalization, the sampling stage begins. Between two consecutive measurements additional Monte Carlo moves are performed in order to reduce correlations between samples. For each sample, the absolute magnetization per spin and the energy per spin are recorded, together with the total magnetization and total energy of the system. From these measurements the average values and their statistical uncertainties are computed using the standard error of the mean.

Finally, the magnetic susceptibility and the heat capacity per spin are calculated using the fluctuation–dissipation relations

$$\chi = \frac{\beta}{N} \left(\langle M^2 \rangle - \langle M \rangle^2 \right)$$

and

$$C_V = \frac{\beta^2}{N} \left(\langle E^2 \rangle - \langle E \rangle^2 \right),$$

where $N = L^2$ is the total number of spins. The resulting quantities are then plotted as functions of the temperature $T = 1/\beta$.

At low temperatures the magnetization is close to one, indicating a strongly ordered ferromagnetic phase in which most spins are aligned. As the temperature increases, the magnetization decreases rapidly and approaches zero at higher temperatures, reflecting the transition to a disordered paramagnetic phase.

The energy per spin approaches the ground-state value $E/N \approx -2$ at low temperatures and increases smoothly with temperature due to thermal fluctuations.

Both the magnetic susceptibility and the heat capacity exhibit pronounced peaks near $T \approx 2.3$, indicating the presence of a phase transition. This value is consistent with the exact critical temperature of the two-dimensional Ising model,

$$T_c = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269$$

for $J = 1$ and $k_B = 1$.
Overall, the numerical results reproduce the expected thermodynamic behavior of the two-dimensional Ising model.

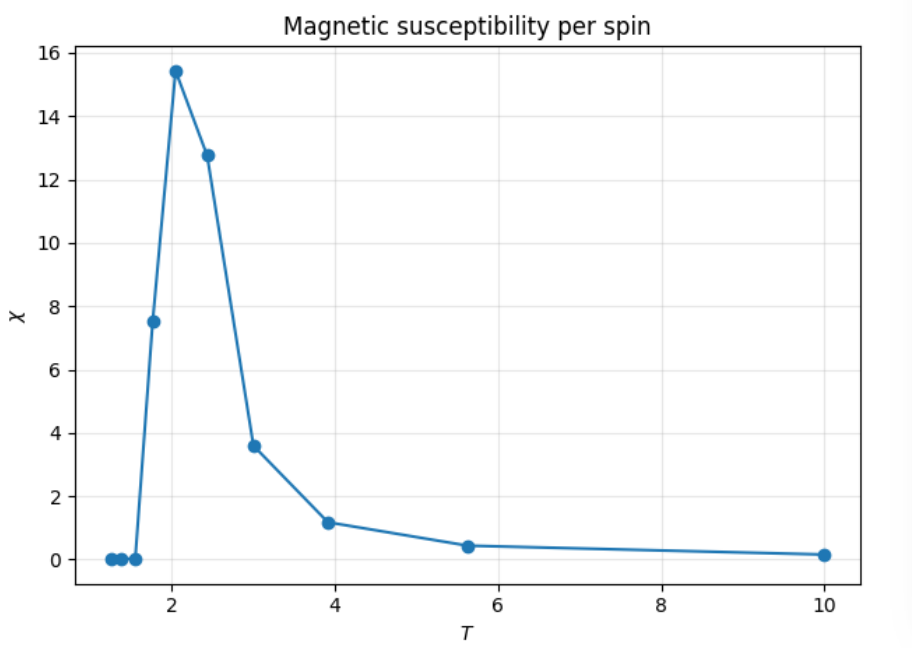

Fig.1 Magnetic susceptibility per spin as function of T.

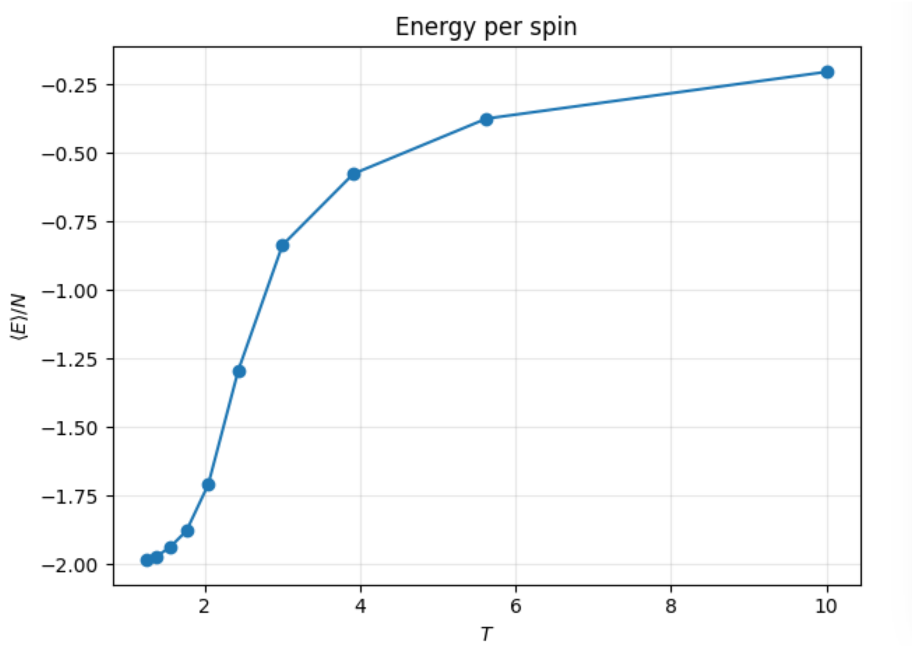

Fig.2 Energy per spin as function of T.

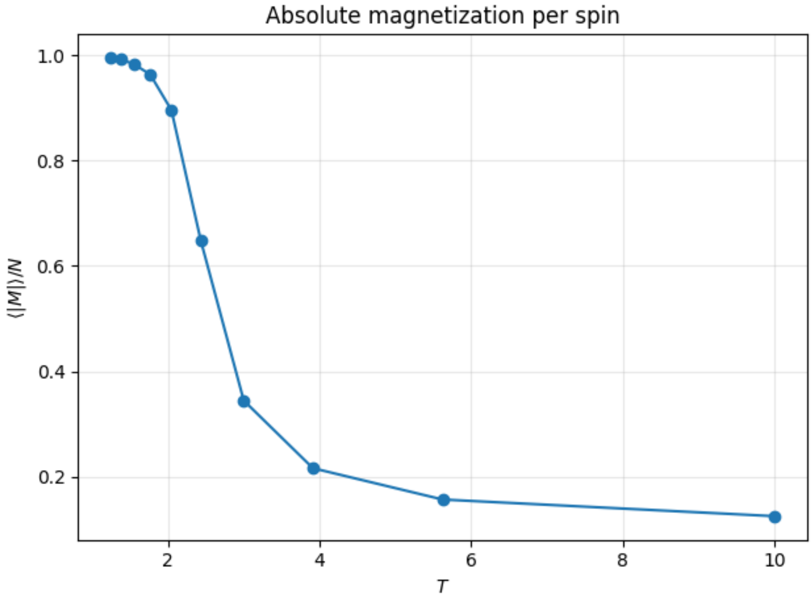

Fig 3. Absolute magetization per spin as function of T.

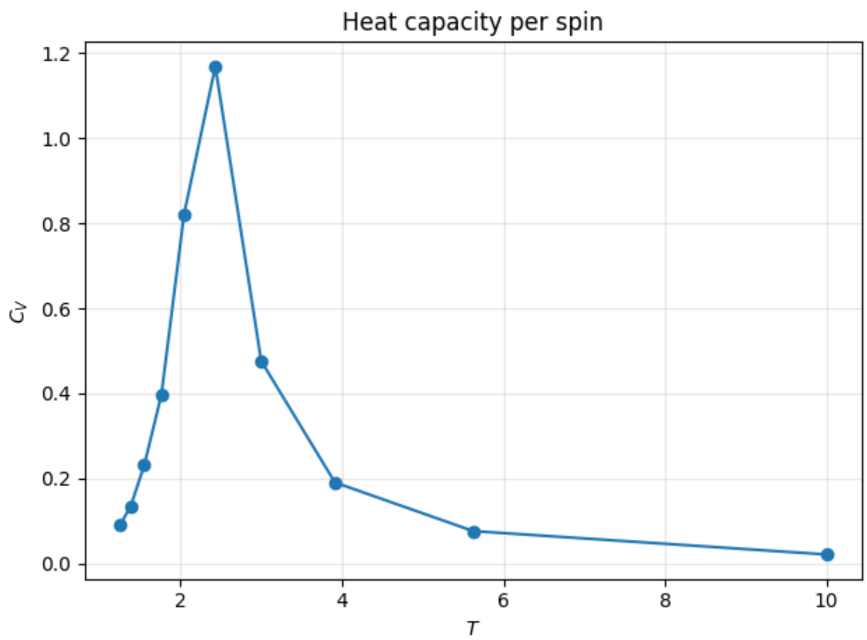

Fig 4. Heat Capacity per spin as function of T.

Error Estimation: Standard Error vs Blocking Method

Two different approaches were considered for estimating the statistical uncertainties of the measured observables. The first approach uses the standard error of the mean, while the second applies a binning analysis in order to account for correlations between Monte Carlo samples.
In the simple approach, the error bars are computed using

$$\sigma_{\bar{Q}} = \frac{\sigma_Q}{\sqrt{N_{\text{sample}}}},$$

where σ_Q is the standard deviation of the measured quantity Q and N_{sample} is the number of samples. This method assumes that the samples are statistically independent. However, in Monte Carlo simulations consecutive configurations are often correlated, which can lead to an underestimation of the true statistical error.
To address this issue, a binning analysis was implemented. In this method the sequence of measurements is divided into blocks of equal size. The average of each block is computed, and the statistical error is then estimated from the fluctuations of the block averages. This procedure effectively reduces the influence of correlations between nearby samples and provides a more reliable estimate of the uncertainty.
In practice, the blocking method typically yields slightly larger error bars than the naive standard error estimate. This reflects the presence of autocorrelations in the Monte Carlo data and leads to a more realistic assessment of the statistical uncertainties.
The binning analysis shows that the statistical uncertainties are small. In particular, for the magnetization the error bars are typically between 10^{-4} and 10^{-3}, reaching values of order 5×10^{-3} near the critical region. As a result, the error bars are not so visible in the plots, especially away from the phase transition, where the fluctuations are much smaller.

Task 2: Determination of the critical temperature

The critical temperature T_c was estimated from the temperature at which the magnetic susceptibility χ and the heat capacity C_V reach their maximum values. These quantities typically exhibit pronounced peaks near the phase transition.
From the numerical results we obtained

$$T_c^{(\chi)} \approx 2.0455$$

from the maximum of the magnetic susceptibility, and

$$T_c^{(C_V)} \approx 2.4324$$

from the maximum of the heat capacity.
The estimated values are reasonably close to the exact result. The small deviations arise mainly from finite-size effects and the finite number of Monte Carlo samples used in the simulation. For finite lattices, the peaks of χ and C_V occur at slightly different temperatures, producing so-called pseudo-critical temperatures $T_c(L)$ that converge to the exact value as the system size increases.

Task 3: Dependence on the system size

To investigate finite-size effects, the Monte Carlo simulation was repeated for different lattice sizes $L = 4, 8, 16$, and 32. For each system size we measured the energy, absolute magnetization, magnetic susceptibility χ, and heat capacity C_V as functions of temperature.
As the lattice size increases, the transition between the ferromagnetic and paramagnetic phases becomes sharper. In particular, the peaks of the susceptibility and heat capacity become more pronounced for larger systems, which is consistent with the behavior expected when approaching the thermodynamic limit.
For finite systems, the positions of these peaks define pseudo-critical temperatures $T_c(L)$. From the numerical results we obtain

$$\begin{aligned} L = 4: &\quad T_c^{(\chi)} \approx 1.5000, \quad T_c^{(C_V)} \approx 2.3205, \\ L = 8: &\quad T_c^{(\chi)} \approx 2.0641, \quad T_c^{(C_V)} \approx 2.3718, \\ L = 16: &\quad T_c^{(\chi)} \approx 2.1667, \quad T_c^{(C_V)} \approx 2.2692, \\ L = 32: &\quad T_c^{(\chi)} \approx 2.3205, \quad T_c^{(C_V)} \approx 2.2179. \end{aligned}$$

From the results we observe that the estimates obtained from the heat capacity are generally closer to the theoretical value, while the susceptibility-based estimates show larger deviations for small lattice sizes. As the system size increases, the pseudo-critical temperatures tend to approach the exact value, although fluctuations and finite-size effects remain visible in the numerical results.

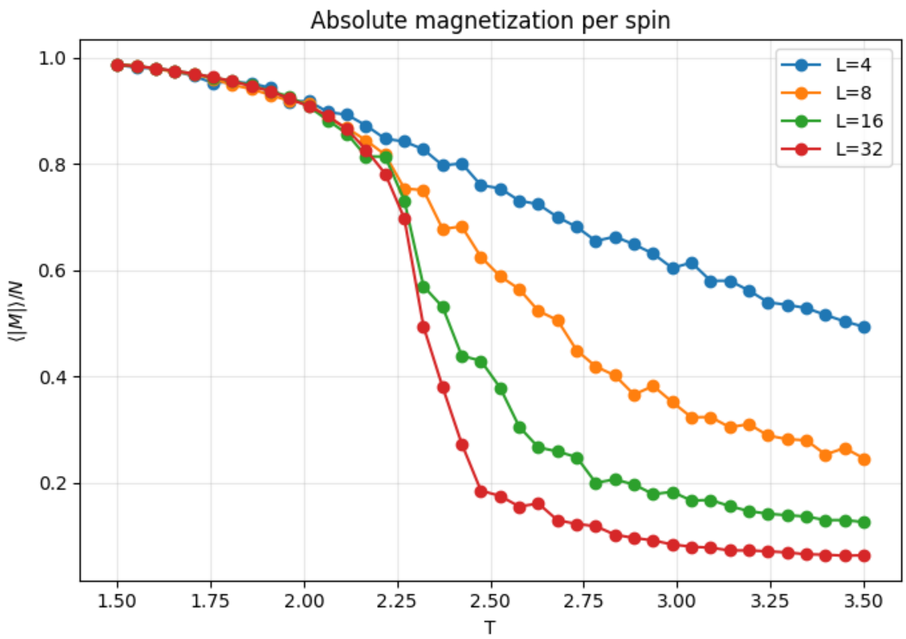

Fig 5. Absolute magnetization per spin as function of T, for L=4,8,16,32.

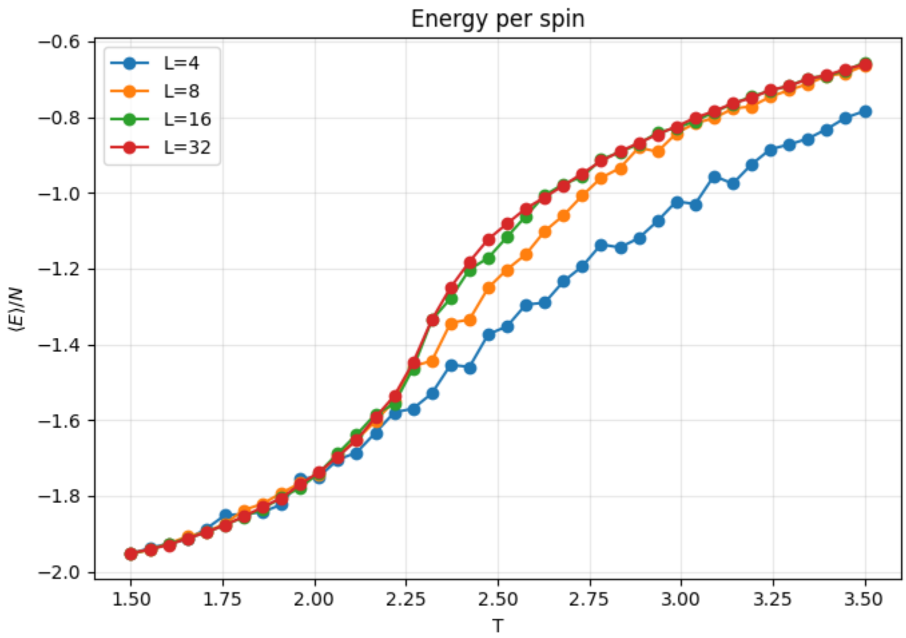

Fig 6. Energy per spin as function of T, for L=4,8,16,32.

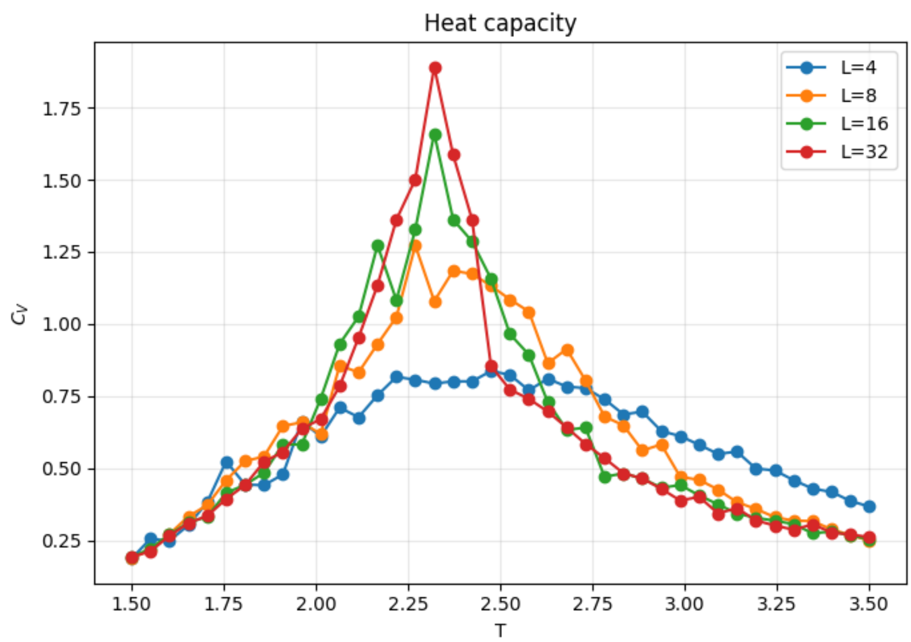

Fig 7. Heat Capacity per spin as function of T, for L=4,8,16,32.

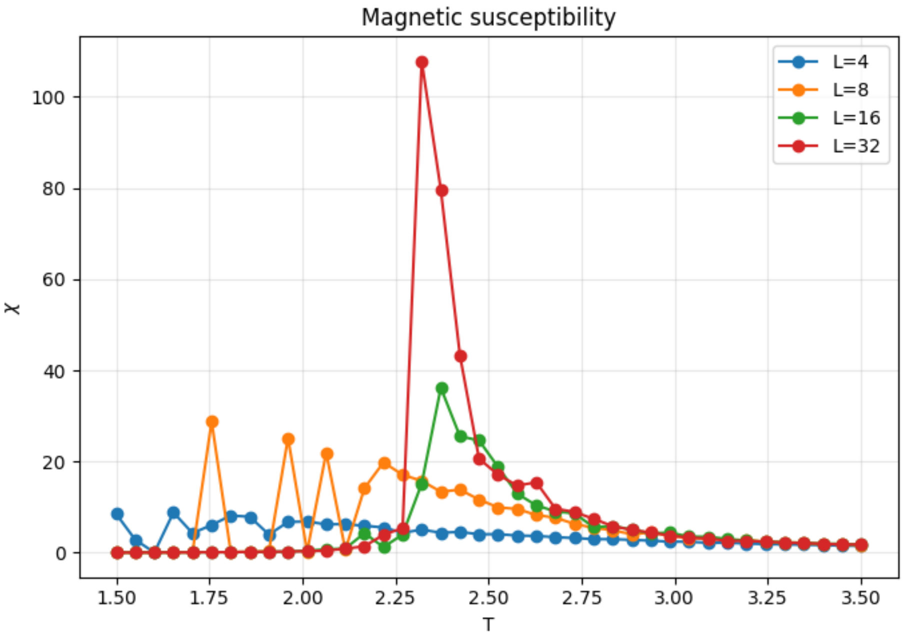

Fig 8. Magnetic susceptibility per spin as function of T, for L=4,8,16,32.

1 Task 4: Optimization Using a Probability Lookup Table

Task 4: Precomputed spin-flip acceptance probabilities To reduce the computation time, the Metropolis acceptance probabilities were precomputed and stored in a lookup table. In the two-dimensional Ising model, the energy difference associated with a single-spin flip can only take the values

$$\Delta E \in \{-8, -4, 0, 4, 8\}$$

for $J = 1$. Therefore, instead of evaluating the exponential function repeatedly during the simulation, the corresponding acceptance probabilities were computed once for each temperature and then reused throughout the Monte Carlo updates.

This optimization does not modify the Metropolis algorithm itself and therefore does not change the physical results. It only improves the computational efficiency of the simulation.

Using this implementation, we obtained the following pseudo-critical temperatures:

$$\begin{aligned} L = 4: &\quad T_c^{(\chi)} \approx 1.5513, \quad T_c^{(Cv)} \approx 2.4231 \\ L = 8: &\quad T_c^{(\chi)} \approx 2.2179, \quad T_c^{(Cv)} \approx 2.4231 \\ L = 16: &\quad T_c^{(\chi)} \approx 2.2692, \quad T_c^{(Cv)} \approx 2.3205 \\ L = 32: &\quad T_c^{(\chi)} \approx 2.3205, \quad T_c^{(Cv)} \approx 2.2692 \end{aligned}$$

The results follow the expected finite-size behavior. For larger lattice sizes the estimated critical temperature is close to the exact value, while smaller systems show stronger finite-size effects and larger fluctuations.

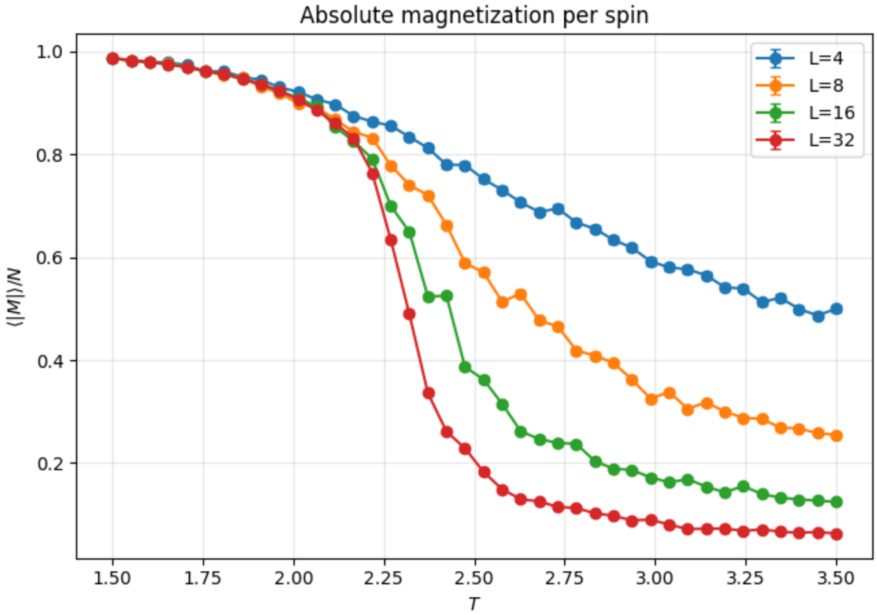

Fig 9. Absolute magnetization per spin as function of T, for L=4,8,16,32, with precomputed probabilities.

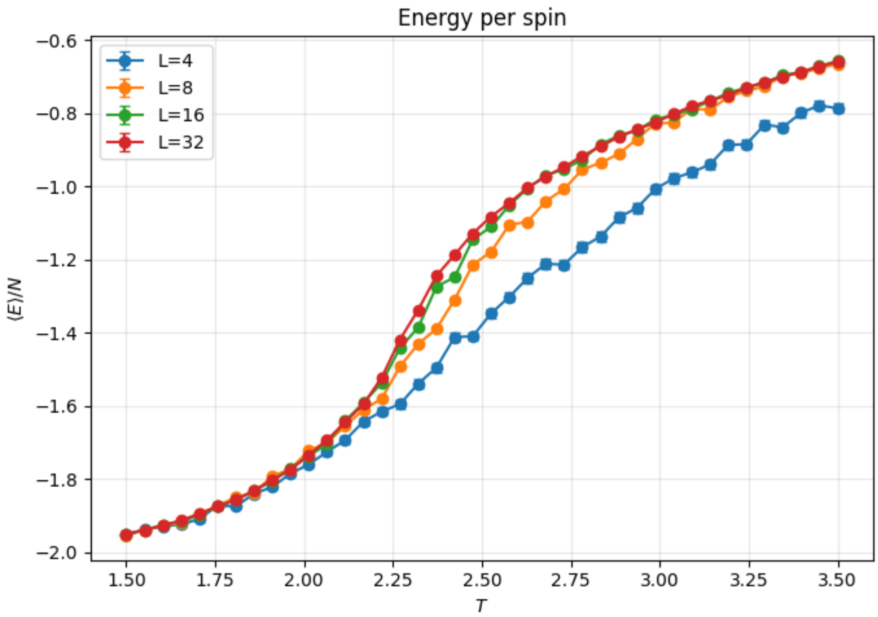

Fig 10. Energy per spin as function of T, for L=4,8,16,32, with precomputed probabilities.

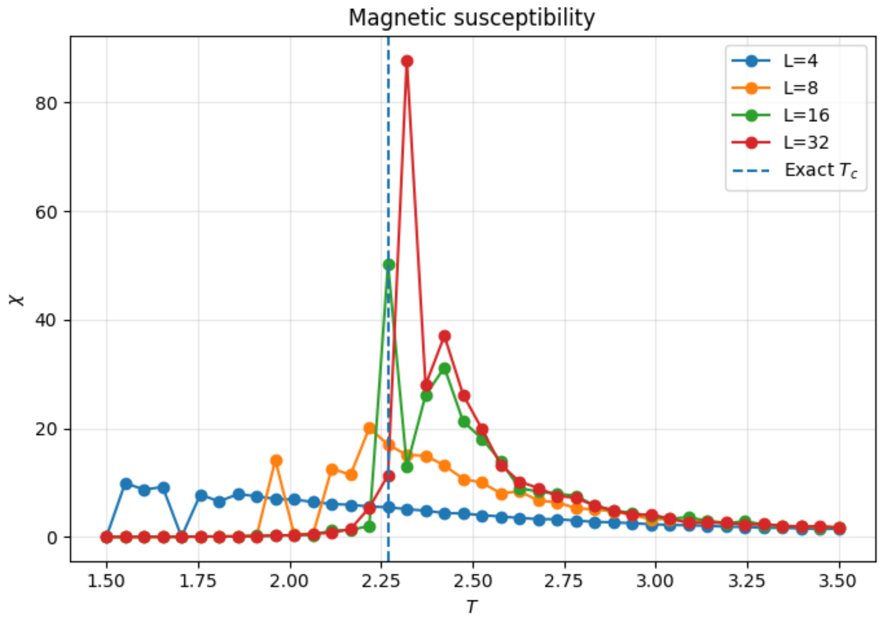

Fig 11. Magnetic susceptibility per spin as function of T, for L=4,8,16,32, with precomputed probabilities.

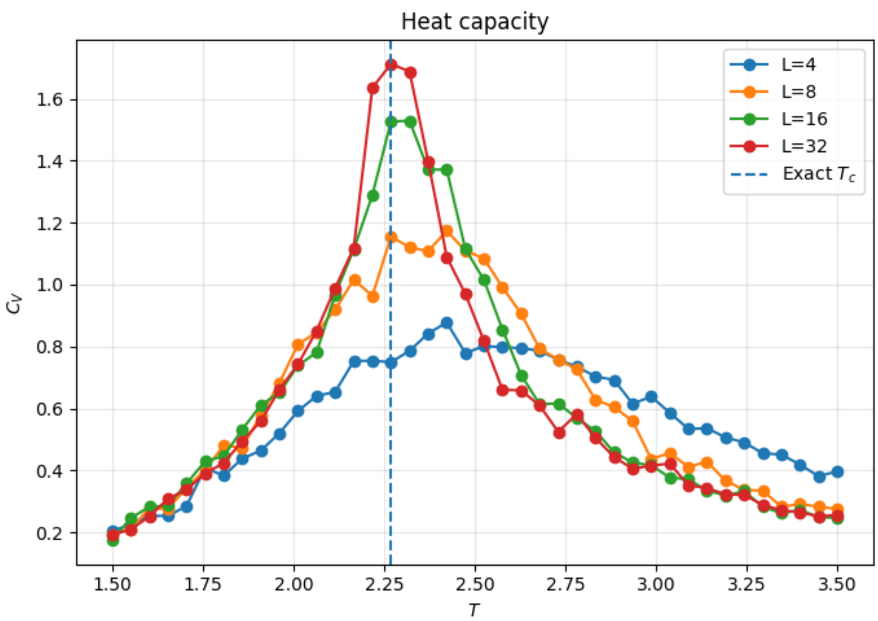

Fig 12. Heat Capacity per spin as function of T, for L=4,8,16,32, with precomputed probabilities.

2 Task 5: Magnetization as a Function of Simulation Time

To study the time dependence of the magnetization, we measured M as a function of Monte Carlo simulation time for a temperature below the critical temperature, using a small lattice in order to make sign flips observable.

In this simulation the time evolution of the magnetization is studied for a two-dimensional Ising lattice with $L = 4$ at temperature $T = 1.5$, below the critical temperature $T_c \approx 2.269$. Starting from a random configuration, the system is first equilibrated using the Metropolis algorithm. After equilibration, the magnetization per spin,

$$\frac{M(t)}{N},$$

is recorded after each Monte Carlo sweep ($N = L^2$ spin updates).
For $T < T_c$ the system is in the ferromagnetic phase and tends to remain in one of two ordered states with positive or negative magnetization. In our simulation, the time series shows that the magnetization remains close to ± 1 for extended periods of time and occasionally switches sign.
These sign flips occur because, for a finite system, thermal fluctuations allow the system to cross the free-energy barrier between the two equivalent ordered phases. Such reversals are therefore a finite-size effect and illustrate how a small system can explore both symmetry-related states during the simulation.

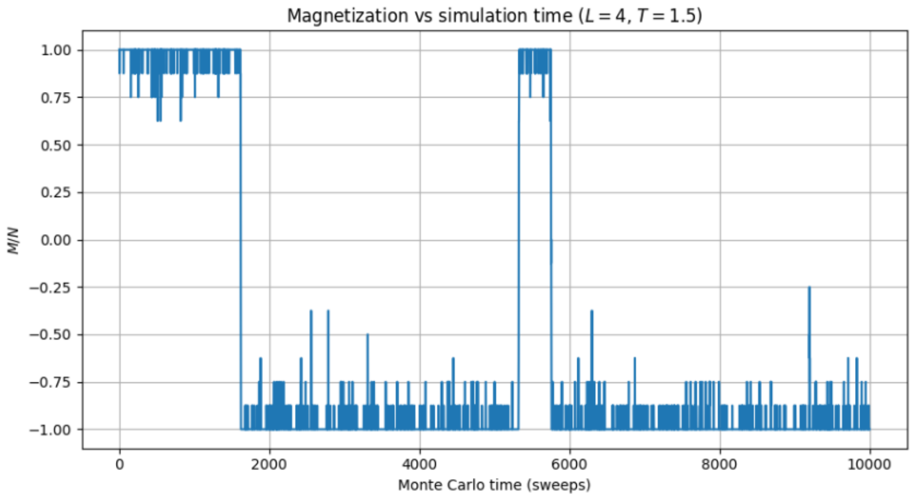

Fig 13. Magnetization vs simulation time (L=4,T=1.5).

References

- Course material from the Computational Statistical Physics lectures, including lecture slides and the provided skeleton Python code for the Monte Carlo simulation of the Ising model.